

Fiscal Policy

ECON 101 - Macroeconomics

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- **12.1 What is Fiscal Policy?**
- 12.2 The Effects of Fiscal Policy on Real GDP and the Price Level
- 12.4 Government Purchases and Tax Multipliers
- 12.5 The Limits of Fiscal Policy as a Stimulus
- 12.6 Deficits, Surpluses, and Federal Government Debt
- 12.7 The Effects of Fiscal Policy in the Long Run
- **Appendix A Closer Look at the Multiplier**

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12.1 Learning Objective

Goal

Define fiscal policy.

- Understand what economists mean by **fiscal policy**.
- Distinguish between:
 - automatic stabilizers, and
 - discretionary fiscal policy.
- Connect fiscal policy to real-world Canadian examples (EI, tax changes, infrastructure spending).

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- **Fiscal policy** refers to changes in **federal taxes and purchases** that are intended to achieve **macroeconomic policy objectives**.
- Provincial and territorial taxes and spending:
 - are important,
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- The key idea:
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Automatic Stabilizers vs. Discretionary Policy

- Some forms of government spending and taxes automatically move with the business cycle:

Automatic stabilizers

Government spending and taxes that **automatically increase or decrease** with the business cycle, without any new legislation.

- Example:
 - Employment Insurance (EI) payments rise in a recession:
 - more people become unemployed and qualify for benefits.
- **Discretionary fiscal policy:**
 - intentional actions the government takes to change spending or taxes,
 - e.g. a new infrastructure program, or a legislated cut in the GST.

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Explain how fiscal policy affects aggregate demand and how the government can use fiscal policy to stabilize the economy.

- The government carries out fiscal policy through:
 - changes in **government purchases** (G),
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- A change in G affects aggregate demand **directly**.
- A change in T affects income and consumption, and so affects aggregate demand **indirectly**.

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Expansionary vs. Contractionary Fiscal Policy

- **Expansionary fiscal policy:**
 - increasing government purchases, or
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- **Contractionary fiscal policy:**
 - decreasing government purchases, or
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- Both types work through the aggregate demand channel:
 - higher G and lower T shift AD right,
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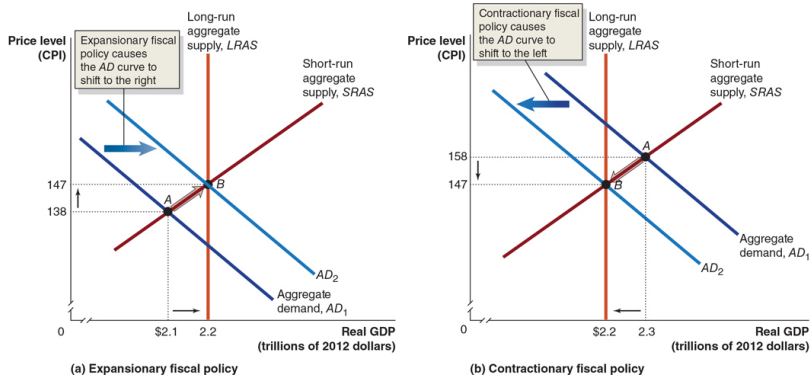
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Figure 12.7 – Fiscal Policy (1 of 2)

Figure 12.7 Fiscal Policy

In panel (a), the economy begins in recession at point A, with real GDP of \$1.75 trillion and a price level of 125. An expansionary fiscal policy will cause aggregate demand to shift to the right, from AD_1 to AD_2 , increasing real GDP from \$1.75 trillion to \$1.77 trillion and the price level from 125 to 127 (point B). In panel (b), the economy begins at point A, with real GDP at \$1.79 trillion and the price level at 127. Because real GDP is greater than potential GDP, the economy will experience rising wages and prices. A contractionary fiscal policy will cause aggregate demand to shift to the left, from AD_1 to AD_2 , decreasing real GDP from \$1.79 trillion to \$1.77 trillion and the price level from 129 to 127 (point B).



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- **Expansionary fiscal policy** involves:
 - increasing government purchases, or
 - decreasing taxes.
- If the government believes real GDP will be **below potential GDP**:
 - it can enact expansionary fiscal policy to close the **recessionary gap**.
- Effects:
 - AD shifts right,
 - real GDP and employment rise,
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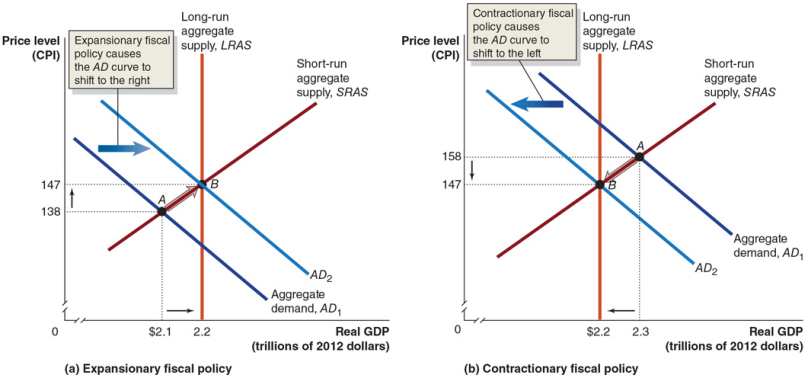
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- **Contractionary fiscal policy** involves:
 - decreasing government purchases, or
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- If the government believes real GDP will be **above potential GDP**:
 - it can enact contractionary policy to reduce inflationary pressure.
- Effects:
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Table 12.1 Countercyclical Fiscal Policy

Problem	Type of Policy Required	Actions by Congress and the President	Result
Recession	Expansionary	Increase government purchases or cut taxes	Real GDP and the price level rise.
Rising inflation	Contractionary	Decrease government purchases or raise taxes	Real GDP and the price level fall.

Interpreting Table 12.1

- The table summarizes **countercyclical** fiscal policy:
 - policy that moves the economy in the opposite direction of the business cycle.
- Bear in mind:
 - These effects are described **ceteris paribus**:
 - everything else (including monetary policy) is assumed unchanged.
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Explain how the government purchases and tax multipliers work.

- An initial change in G or T sets off:
 - an **autonomous** change in spending, and
 - a chain of **induced** changes via consumption.
- Together these create a **multiplier effect** on real GDP.

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- People receiving this extra spending see their income rise:
 - they increase consumption spending,
 - which further raises AD – this is the **induced** increase.
- The series of induced increases in consumption spending that results from the initial increase in autonomous expenditures is the:

Multiplier effect

The process where an initial change in autonomous spending leads to a larger change in equilibrium real GDP.

Autonomous vs. Induced Effects and the Multiplier

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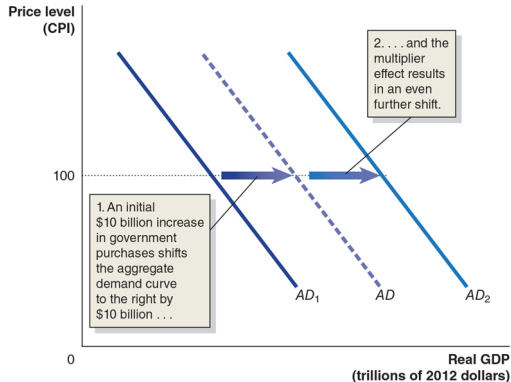
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Figure 12.10 – The Multiplier Effect and AD

Figure 12.10 The Multiplier Effect and Aggregate Demand

An initial increase in government purchases of \$10 billion causes the aggregate demand curve to shift to the right, from AD_1 to the dashed AD curve, and represents the effect of the initial increase of \$10 billion in government purchases. Because this initial increase raises incomes and leads to further increases in consumption spending, the aggregate demand curve will ultimately shift further to the right, to AD_2 .



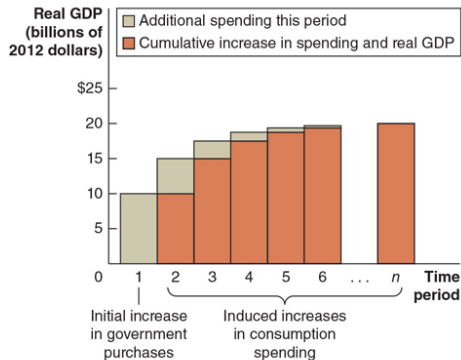
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Figure 12.11 – Example: \$10 Billion Increase in G

Figure 12.11 The Multiplier Effect of an Increase in Government Purchases

Following an initial increase in government purchases, spending and real GDP increase over a number of periods due to the multiplier effect. The new spending and increased real GDP in each period is shown in green, and the level of spending from the previous period is shown in orange. The sum of the orange and green areas represents the cumulative increase in spending and real GDP. In total, equilibrium real GDP will increase by \$20 billion as a result of an initial increase of \$10 billion in government purchases.

Time Period	Additional Spending This Period	Cumulative Increase in Spending and Real GDP
1	\$10 billion in government purchases	\$10 billion
2	\$5 billion in consumption spending	\$15 billion
3	\$2.5 billion in consumption spending	\$17.5 billion
4	\$1.25 billion in consumption spending	\$18.75 billion
5	\$0.625 billion in consumption spending	\$19.375 billion
6	\$0.3125 billion in consumption spending	\$19.6875 billion
⋮	⋮	⋮
n	0	\$20 billion



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Numerical Example of the Multiplier

- Suppose each increase in spending induces an increase in consumption equal to one-half of the increase in income:
 - that is, the marginal propensity to consume (MPC) on domestic goods is 0.5.
- A \$10 billion increase in government purchases:
 - initially raises AD by \$10 billion (autonomous),
 - then induces an additional \$5 billion in new consumption,
 - then another \$2.5 billion, and so on.
- Over time, the total increase in real GDP is **larger** than the initial \$10 billion increase in G .

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Multipliers for Government Purchases and Taxes

- We describe the total effect of a change in G or T by the change in equilibrium real GDP:

- Government purchases multiplier:

$$\frac{\Delta Y}{\Delta G}$$

- Tax multiplier:

$$\frac{\Delta Y}{\Delta T}$$

- The tax multiplier is **negative**:
 - an increase in taxes decreases equilibrium real GDP,
 - a decrease in taxes increases equilibrium real GDP.
- The tax multiplier is **smaller in absolute value** than the government purchases multiplier, because a tax cut is partially *saved* and partially *spent*.

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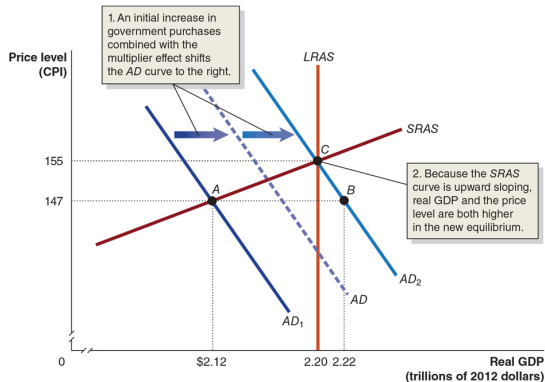
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Figure 12.12 – Multiplier and AS (Placeholder)

Figure 12.12 The Multiplier Effect and Aggregate Supply

The economy is initially at point A. An increase in government purchases causes the aggregate demand curve to shift to the right, from AD_1 to the dashed AD curve. The multiplier effect results in the aggregate demand curve shifting further to the right, to AD_2 (point B). Because of the upward-sloping supply curve, the shift in aggregate demand results in a higher price level. In the new equilibrium at point C, both real GDP and the price level have increased. The increase in real GDP is less than indicated by the multiplier effect with a constant price level.



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Multipliers with an Upward-Sloping SRAS

- With an upward-sloping short-run aggregate supply (SRAS) curve:
 - an increase in AD raises **both** real GDP and the price level.
- Example:
 - suppose autonomous and induced effects increase AD by \$100 billion.
 - Real GDP might increase from \$2,120 billion to \$2,220 billion,
 - while the price level also rises.
- So part of the multiplier “shows up” as higher prices, not just higher real output.

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Multipliers Work in Both Directions

- An **increase** in government purchases and a **cut** in taxes have a **positive** multiplier effect.
- A **decrease** in government purchases and an **increase** in taxes have a **negative** multiplier effect.
- Example:
 - a reduction in government spending on infrastructure initially affects contractors,
 - then suppliers and employees of those contractors,
 - then other firms and workers,
 - leading to a multiplied reduction in real GDP.

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12.5 Learning Objective

Goal

Discuss the difficulties that can arise in implementing fiscal policy.

- Fiscal policy is powerful in theory but may be less effective in practice.
- Key issues:
 - timing and lags,
 - crowding out of private spending.

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 - Parliament must debate and agree on changes to spending or taxes.
 2. **Implementation delay**
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Crowding out

A decline in private expenditures (consumption, investment, or net exports) as a result of an increase in government purchases.

- When the government increases spending, it often finances this by issuing bonds.
- Higher government borrowing can:
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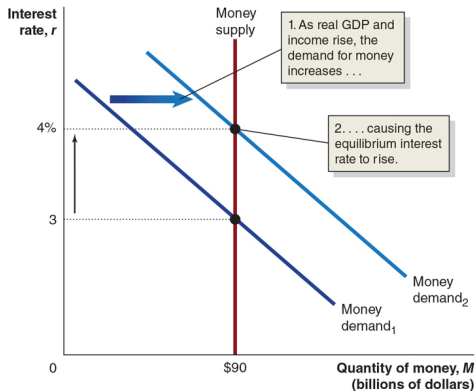
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Figure 12.13 – Expansionary Fiscal Policy and Interest Rates

Figure 12.13 An Expansionary Fiscal Policy Increases Interest Rates

If the federal government increases spending, the demand for money will increase from Money demand₁ to Money demand₂ as real GDP and income rise. With the supply of money constant at \$90 billion, the result is an increase in the equilibrium interest rate from 3 percent to 4 percent, which crowds out some consumption, investment, and net exports.

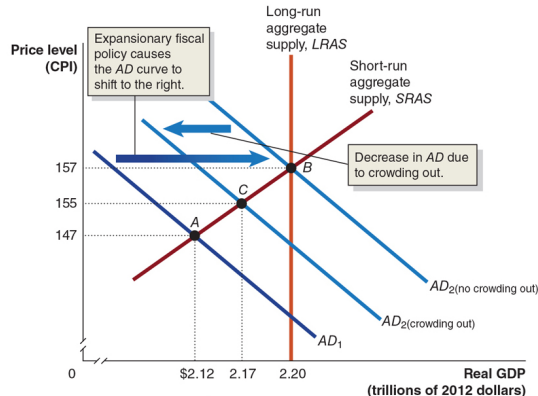


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Figure 12.14 – Crowding Out in the Short Run

Figure 12.14 The Effect of Crowding Out in the Short Run

The economy begins in a recession, with real GDP of \$2.12 trillion (point A). In the absence of crowding out, an increase in government purchases will shift aggregate demand to $AD_{2(\text{no crowding out})}$ and bring the economy to equilibrium at potential GDP of \$2.20 trillion (point B). But the higher interest rate resulting from the increased government purchases will reduce consumption, investment, and net exports, causing aggregate demand to shift to $AD_{2(\text{crowding out})}$. The result is a new short-run equilibrium at point C, with real GDP of \$2.17 trillion, which is \$30 billion short of potential GDP.



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Crowding Out: Short Run vs. Long Run

- In the **short run**:
 - an increase in G raises AD,
 - but higher interest rates partly offset this via lower C , I , and NX .
- In the **long run**:
 - the economy returns to potential GDP,
 - the reduction in private spending ultimately **exactly offsets** the increase in government purchases,
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Goal

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- Clarify what we mean by the federal budget deficit and debt.
- See how the budget automatically helps stabilize the economy.

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- A **budget deficit** occurs when:
 - government expenditures are **greater** than tax revenues.
- A **budget surplus** occurs when:
 - government expenditures are **less** than tax revenues.
- Question for students:
 - Is the federal government currently running a deficit or a surplus?

- A **budget deficit** occurs when:
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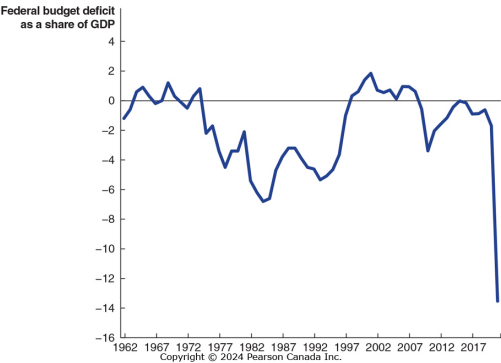
Figure 12.15 – Federal Budget Deficit, 1962–2021 (Placeholder)

Figure 12.15 The Federal Budget Deficit, 1962–2021

The federal government has run a budget deficit for most years since 1962. The federal budget balance tends to be more in deficit when the economy is in recession, and the deficit tends to be smaller when the economy is doing well. You should note that the federal government has changed its accounting standards at least twice over this time period, which makes comparisons between years that are not close together difficult.

NOTE: In this graph, the years represent the end of the fiscal year. For example, the fiscal year 2017–2018 is identified as 2018.

SOURCE: Government of Canada, Department of Finance, "Fiscal Reference Tables for 2021," <https://www.canada.ca/en/departement-finance/services/publications/fiscal-reference-tables/2021.html>, and for 1996, <https://www.canada.ca/content/dam/fin/migration/frt-trf/pdf/frt96e.pdf>.



Deficits Over the Business Cycle

- The federal government has run deficits in most years since the 1960s.
- Deficits tend to be **larger in recessions**:
 - tax revenues fall as incomes and profits decline,
 - automatic stabilizers like EI and welfare spending rise.
- These automatic stabilizers help:
 - limit the severity of recessions,
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- When the federal government runs a deficit, it finances this by selling **bonds**.
- The total value of outstanding federal government bonds is:

Federal government debt (national debt)

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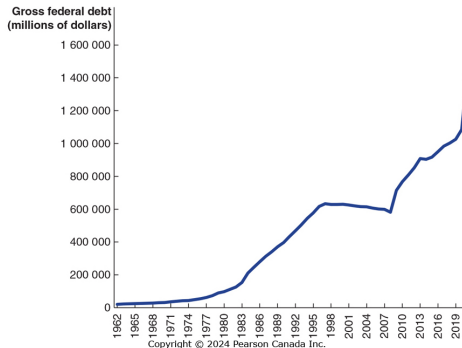
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Figure 12.16 – Federal Government Debt, 1962–2021

Figure 12.16 The Federal Government Debt, 1962–2021

The federal government debt increases whenever the federal government runs a budget deficit. The large deficits of the 1980s and 1990s increased the debt significantly. The debt actually began to fall in the 2000s, as the federal government brought deficits down and began to run surpluses. This trend continued until 2009, when the federal government engaged in expansionary fiscal policy to offset the global recession. The debt stabilized again in 2013–2014 but began rising in 2015. In 2020 to 2022, the federal debt rose dramatically as government revenue initially fell and expenditures rose rapidly.

SOURCE: Government of Canada, Department of Finance, "Fiscal Reference Table for 2021," <https://www.canada.ca/en/departement-finance/services/publications/fiscal-reference-tables/2021.html>, and for 1996, <https://www.canada.ca/content/dam/fin/migration/frt-trf/pdf/frt96e.pdf>.



Is Government Debt a Problem?

- Currently, the federal government faces no serious risk of default because:
 - it can borrow at relatively low interest rates,
 - interest payments are small relative to the overall budget.
- In the **long run**, a debt that grows faster than GDP can be problematic:
 - it may crowd out private investment,
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12.7 Learning Objective

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Discuss the effects of fiscal policy in the long run.

- So far, we focused on short-run stabilization (AD).
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- Some fiscal policy actions aim at long-run increases in potential GDP:
 - these affect **aggregate supply**, not just aggregate demand.
- Policies that focus on incentives to work, save, invest, and innovate are often called:

Supply-side economics

Fiscal policy aimed at increasing long-run aggregate supply, usually by changing taxes or regulations.

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Tax Wedge and Marginal Tax Rates

- Most taxes are assessed as a percentage of economic activity:
 - individual income, corporate profits, capital gains, etc.
- When an individual decides how much to work, they care about the:
 - **post-tax wage**: how much an extra hour of work increases their consumption possibilities.
- A firm deciding how many workers to hire cares about:
 - the **pre-tax wage**: the total cost of employing each worker.
- The difference between pre-tax and post-tax returns is known as the:

Tax wedge

The gap between the pre-tax and post-tax return to an economic activity, implied by the *marginal tax rate*.

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Why Tax Rates Matter

- The larger the tax wedge, the stronger the behavioural response:
 - individuals and firms have weaker incentives to engage in taxed activities.
- Examples:
 - **Individual income tax:**
 - affects labour supply decisions and entrepreneurship.
 - **Corporate income tax:**
 - affects incentives to undertake investment projects.
 - **Taxes on dividends and capital gains:**
 - affect the supply of loanable funds.
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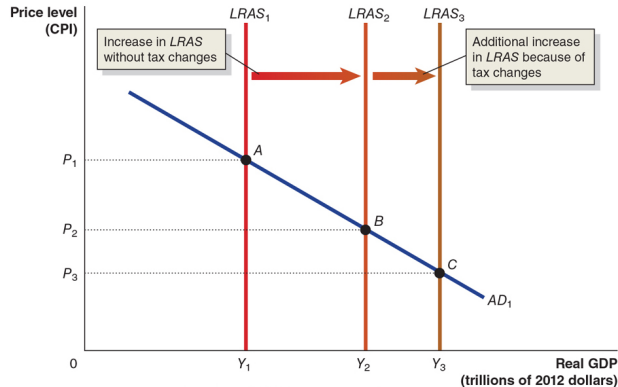
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Figure 12.17 – Supply-Side Effects of a Tax Change

Figure 12.17 The Supply-Side Effects of a Tax Change

The economy's initial equilibrium is at point A. With no tax change, the long-run aggregate supply curve shifts to the right, from $LRAS_1$ to $LRAS_2$. Equilibrium moves to point B, with the price level falling from P_1 to P_2 and real GDP increasing from Y_1 to Y_2 . With tax reductions and simplifications, the long-run aggregate supply curve shifts further to the right, to $LRAS_3$, and equilibrium moves to point C, with the price level falling to P_3 and real GDP increasing to Y_3 .



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- Tax reform has the potential to increase real GDP in the long run:
 - by increasing work effort,
 - by raising saving and investment,
 - by encouraging entrepreneurship and innovation.
- But the **magnitude** of these effects is uncertain:
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- Economists agree that more research is needed on the quantitative impact of tax reforms.

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Apply the multiplier formula using a simple macroeconomic model.

- Build a simple model of how real GDP is determined.
- Derive:
 - the government purchases multiplier,
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An Expression for Equilibrium Real GDP (1 of 2)

- For simplicity, initially assume:
 - taxes do not depend on income (fixed amount),
 - there are no government transfers to households,
 - there are no imports or exports (closed economy),
 - the price level is constant.
- Consider the following system (numbers in billions):

Equation	Interpretation
1. $C = 1,000 + 0.8(Y - T)$	Consumption function
2. $I = 125$	Planned investment function
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1. $C = 1,000 + 0.8(Y - T)$	Consumption function
2. $I = 125$	Planned investment function
3. $G = 125$	Government purchases function
4. $T = 100$	Tax function
5. $Y = C + I + G$	Equilibrium condition

An Expression for Equilibrium Real GDP (1 of 2)

- For simplicity, initially assume:
 - taxes do not depend on income (fixed amount),
 - there are no government transfers to households,
 - there are no imports or exports (closed economy),
 - the price level is constant.
- Consider the following system (numbers in billions):

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An Expression for Equilibrium Real GDP (2 of 2)

- These numbers are in billions of dollars.
- Consumption depends on disposable income $Y - T$.
- Substituting into the equilibrium condition $Y = C + I + G$, the slides write:

$$Y = 100 + 0.8(Y - 100) + 125 + 125$$

- Subtracting $0.8Y$ from both sides and simplifying leads to an equilibrium value of real GDP of approximately \$13 trillion in this example.
- The exact numbers are less important than the **method**:
 - substitute behavioural equations into the equilibrium condition,
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Algebraic Expression for Equilibrium Real GDP

- Instead of plugging in numbers, we can work algebraically.

- Let:

$$C = \bar{C} + \text{MPC}(Y - T), \quad I = \bar{I}, \quad G = \bar{G}, \quad T = \bar{T}$$

- The equilibrium condition:

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Government Purchases Multiplier

- Any change in an autonomous variable changes equilibrium Y .
- Write the equilibrium condition in *change* form (using Δ):

$$\Delta Y = \text{MPC} \Delta Y + \Delta G + \Delta I + \Delta C_{\text{aut}} - \text{MPC} \Delta T$$

- Holding $\Delta C_{\text{aut}} = 0$, $\Delta T = 0$, and $\Delta I = 0$, we get:

$$\Delta Y = \text{MPC} \Delta Y + \Delta G.$$

- Rearranging:

$$\Delta Y(1 - \text{MPC}) = \Delta G \quad \Rightarrow \quad \frac{\Delta Y}{\Delta G} = \frac{1}{1 - \text{MPC}}.$$

- This is the **government purchases multiplier**.

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Numerical Government Purchases Multiplier

- Example: $MPC = 0.8$.

- Then:

$$\text{Government purchases multiplier} = \frac{1}{1 - 0.8} = \frac{1}{0.2} = 5.$$

- Interpretation:

- A \$10 billion increase in government purchases raises equilibrium real GDP by \$50 billion in this simple model.

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- Again, consider the change form, but now allow $\Delta T \neq 0$ and hold $\Delta G = \Delta I = \Delta C_{\text{aut}} = 0$:

$$\Delta Y = \text{MPC}(\Delta Y - \Delta T).$$

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$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}}.$$

- Example: if $\text{MPC} = 0.8$,

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Tax Multiplier: Numerical Example

- With $MPC = 0.8$:
 - a \$10 billion *increase* in taxes decreases real GDP by \$40 billion,
 - a \$10 billion *cut* in taxes increases real GDP by \$40 billion.
- Note:
 - the tax multiplier is **negative**,
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The Balanced Budget Multiplier

- Suppose the government increases G and T by the **same** amount, say \$10 billion.
- The total effect on real GDP combines:
 - the government purchases multiplier (+5 in the example),
 - and the tax multiplier (−4 in the example).

- Adding these:

$$\Delta Y = (+5 - 4) \times 10 \text{ billion} = 1 \times 10 \text{ billion.}$$

- Therefore, the **balanced-budget multiplier** is:

$$\frac{\Delta Y}{\Delta G} = 1$$

when $\Delta G = \Delta T$.

- Equal dollar increases in government spending and taxes lead to the same dollar increase in real GDP in the short run.

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Summary

- Fiscal policy refers to changes in federal taxes and purchases intended to achieve macroeconomic objectives.
- Expansionary fiscal policy (higher G or lower T) is used to close recessionary gaps; contractionary fiscal policy (lower G or higher T) is used to reduce inflationary pressures.
- Government purchases and tax changes have multiplied effects on real GDP through the induced changes in consumption.
- Fiscal stimulus faces timing problems and possible crowding out of private spending.
- The federal budget deficit and debt evolve with the business cycle and policy choices; debt is less problematic when it finances productive investment.
- In the long run, fiscal policy can affect potential GDP through supply-side channels, especially via tax rates and incentives.
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